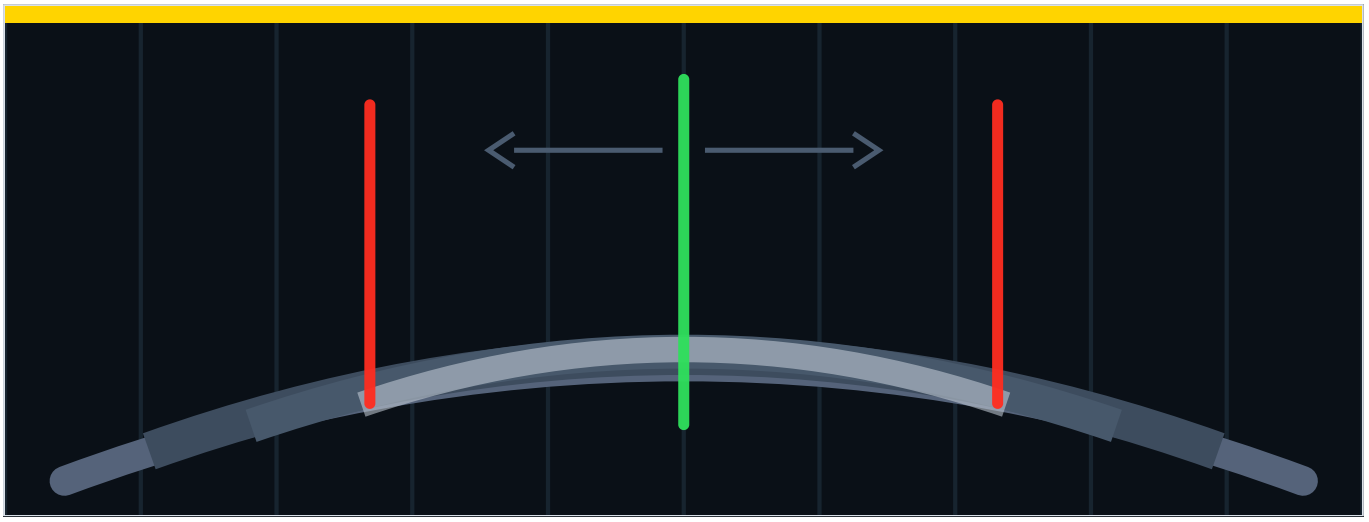
APPLICATION · RUBBER AND COMPOSITE

EDGE AND SYMMETRY REFERENCE IN REINFORCEMENT LAYUP

Reinforcement ply layup width changes from product to product. Every changeover means re-establishing the edge position and the symmetry axis, and setting them by eye makes the result depend on the operator and on the time available. ServoMotion reads the target positions from the recipe and projects the laser reference lines onto the working surface.



Core	K4 dual channel	1,600 mm
SYSTEM USED IN THE EXAMPLE INSTALLATION	1 FIXED + 2 MOVING	TRAVEL

ON THE LINE — DO YOU RECOGNISE THESE SYMPTOMS?

Re-alignment on every product

When the layup width changes, the edge and symmetry reference is rebuilt from scratch, making setup the longest item in the changeover.

Judgement by eye on curved surfaces

On a cylindrical or curved surface, centring the ply depends on the operator’s viewing angle. The same product is set up differently across shifts, resulting in inconsistent output.

The error shows up downstream

Ply misalignment goes unnoticed during layup, and by the time it appears at the next stage, correcting it is expensive.

The reference drifts mechanically

Fixed guides and templates shift with vibration and contact, and need periodic readjustment.

WHAT WE MEASURE DURING THE SITE SURVEY

Layup width range	The difference between the narrowest and widest product determines the housing length and the travel range.
Symmetry axis	Whether the centre reference must remain visible continuously determines the channel configuration.
Surface geometry	Flat, curved or cylindrical geometry affects the projection angle and the mounting height.
Working distance	The vertical distance between the laser and the surface determines line width and angular tolerance.
Support structure	An existing frame or a new traverse; rigidity translates directly into accuracy.
Vibration and environment	Line vibration, surface colour and overhead lighting all influence the choice of laser colour.

WHICH ARCHITECTURE FITS THIS SEGMENT?

MOST APPLICATIONS

CORE PRECISOR

In reinforcement layup, the number of simultaneous reference lines is usually limited. An enclosed industrial profile drives to the target positions under servo control, and where symmetry is required, a dual-channel configuration keeps the centre reference continuously visible.

IF THE NUMBER OF PROJECTED LASER LINES GROWS

SHUTTLE PRECISOR

If more than six lines must be visible at once, the shuttle architecture — a single servo axis positioning multiple laser carriages — takes over. The decision is based on the densest recipe in your portfolio.

EXAMPLE INSTALLATION

CORE PRECISOR · K4 DUAL CHANNEL · 1,600 MM TRAVEL

Two independent motion channels plus a fixed centre reference on a separate bracket. In symmetrical layup, both edges are positioned simultaneously while the centre line stays visible throughout production. These figures come from one example installation; the dimensions for your line are determined during the site survey.

This note summarises the typical requirements of the segment and one example installation. The final system architecture, housing size and scope of connection are established during the site survey. The lasers provide an alignment reference; material and machine elements are set by the operator. Values may change without notice as the products are developed.

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